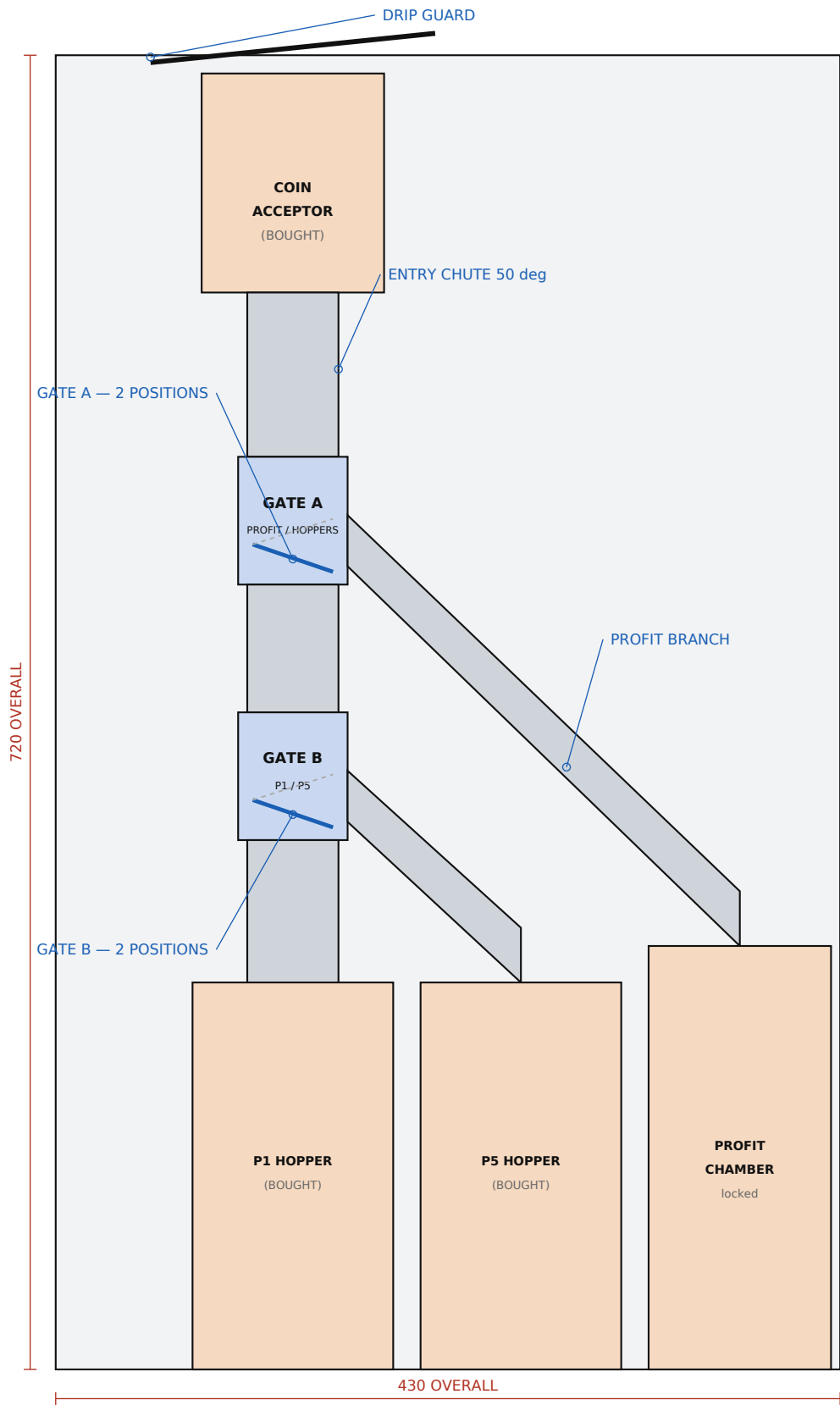
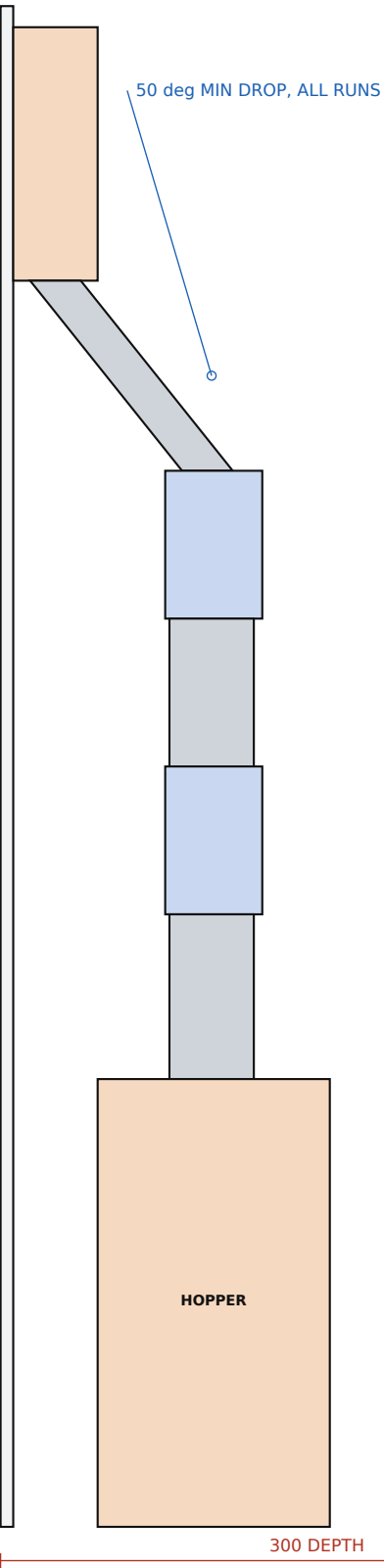


FRONT ELEVATION



SIDE ELEVATION



WHY TWO GATES INSTEAD OF ONE SWING

A binary flap has a HARD MECHANICAL STOP at each end. It either arrives or it visibly does not. A three-position servo has stops only at the extremes; the middle position depends on angular accuracy holding over thousands of cycles against coin impact.

That middle position is where a swing diverter fails, and it fails by dropping PROFIT coins into a CHANGE hopper — corrupting inventory in the direction that costs money.

Cost of the change: one extra micro servo (approx P350) and one pin.
Firmware gets simpler: two booleans instead of a held angle.

EMATHRIX TECHNOLOGIES
Katapatan Ave., Banay-Banay, Cabuyao City, Laguna | DTI 7114652

GENERAL ARRANGEMENT — TWO-GATE DIVERTER
EMX-2026-WATERVENDO-01 COIN ROUTING REV B

ALL DIMENSIONS IN MILLIMETRES

SHEET 1 OF 4
NOT TO SCALE

1 CUT AND CHECK THE COINS FIRST

Before any metal is cut, measure real coins with a caliper: P1, P5, P10, P20, both old BSP and new NGC series. Record diameter and thickness.

Confirm the largest is 30.0 (P20 NGC). If a larger coin exists, every chute dimension on Sheet 1 of Rev A changes. Do not skip this.

2 CHUTE BLANKS

1.5 stainless sheet. Cut blanks for every straight run: entry, gate-to-gate, and the three branches. Fold to a 45 x 6 internal box section, 3 wall.

Fold, do not weld the long seams. A weld bead inside the coin path is a jam.

Deburr and polish every internal edge before assembly, not after.

3 GATE HOUSINGS x2

Both gates are identical — make two of the same part (Sheet 2).

60 x 90 internal box with a 45 x 6 inlet at top, one outlet straight down, one outlet on the side. Fold from one blank where possible.

Drill the pivot hole 4.2 for a 4 dia rod, both side walls, before folding.

4 FLAPS AND PIVOTS

Flap: 50 x 44 x 1.5 stainless. Round the leading edge — a square edge catches a coin arriving mid-travel.

Pivot: 4 dia stainless rod, running in the drilled side walls. No bearing needed. Retain with a split pin or a tack weld at ONE end only.

Crank arm: 20 long, welded to the pivot rod outside the housing.

5 HARD STOPS

Weld or rivet a 6 x 6 stop tab at each end of the flap travel, INSIDE the housing. The stop takes the impact, never the servo gears.

Set the stops so the flap fully closes each outlet with a 1 to 2 overlap.

This is the single most important step for reliability. Do not omit it.

6 SERVO MOUNTING

SG90 micro servo on an external bracket, OUTSIDE the coin path.

Nothing electrical inside the chute. Pushrod: 2 dia stainless with a Z-bend or clevis at both ends.

Set travel 3 to 5 deg beyond each hard stop so the linkage stays loaded and cannot rattle. Record both angles per gate — they go into config.h.

7 ASSEMBLY AND DROP TEST

Assemble entry chute, Gate A, link chute, Gate B and the three branches onto the back panel. Verify 50 deg minimum drop on every run with a digital angle gauge, not by eye.

DROP TEST before wiring: with both flaps set by hand, drop 20 of each coin through each path. Every coin must arrive. Any coin that hangs up is a burr or an angle problem — fix it now, not after wiring.

8 WET TEST

Repeat the drop test with coins deliberately wetted.

Wet coins stick to each other and to the walls. If the wet test fails, increase the drop angle before changing anything else.

Fit the drip guard and re-test. Sign off both tests in writing.

ACCEPTANCE — the assembly is not finished until all four are signed

- ☐ 20 of each coin, each path, DRY — 100% arrival
- ☐ 20 of each coin, each path, WET — 100% arrival
- ☐ Both gates verified against hard stops, angles recorded
- ☐ Service door reaches both gates and both hopper throats

CUT LIST

ITEM	PART	MATERIAL	SIZE	QTY
1	Back mounting panel	Alu 3mm / GI powder-coated	340 x 720	1
2	Entry chute	St.st. 1.5 folded	45 x 6 int. x 115 lg	1
3	Gate housing	St.st. 1.5 folded	60 x 90 x 60 int.	2
4	Gate flap	St.st. 1.5	50 x 44	2
5	Pivot rod	St.st. round	4 dia x 70	2
6	Crank arm	St.st. flat	20 x 8 x 2	2
7	Hard stop tab	St.st. 1.5	6 x 6	4
8	Pushrod	St.st. wire	2 dia x 60	2
9	Servo bracket	Alu 2mm folded	to suit SG90	2
10	Link chute (A to B)	St.st. 1.5 folded	45 x 6 int. x 70 lg	1
11	Branch chute — profit	St.st. 1.5 folded	45 x 6 int. x 150 lg	1
12	Branch chute — hoppers	St.st. 1.5 folded	45 x 6 int. x 140 lg	2
13	Drip guard canopy	St.st. 1.5	160 x 80	1
14	IR sensor bracket	Alu 2mm	to suit hopper outlet	2

MATERIAL NOTES

Do NOT use galvanised sheet anywhere on the coin path. The zinc coating burrs and grabs coins. It is the cheap sheet a shop reaches for by default — state this on the purchase order. Stainless 1.5 is specified for corrosion resistance. This assembly shares a cabinet with a chilled tank, a pump and a dispensing nozzle. Condensation is constant. All internal coin-contact surfaces deburred and polished before assembly. A single burr is a permanent jam point that no amount of firmware will fix. No fastener head, weld bead or bracket may protrude into the coin path. Fold seams outward. Fold rather than weld the chute long seams wherever the geometry allows.

WHY THE COIN PATH DOES NOT SORT BY SIZE

Old BSP P5 is 27.0 dia. New NGC P10 is also 27.0 dia. Identical. Old P10 at 26.5 is half a millimetre from old P5. No chute geometry can separate these — diameter alone cannot tell a P5 from a P10 in this currency.

The chute is therefore a dumb pipe. The ACCEPTOR does the identification, using diameter, thickness AND an eddy-current coil that reads the alloy. Old P5 is nickel-brass; new P10 is bimetallic. Same diameter, completely different electromagnetic signature.

ACCEPTOR PROGRAMMING — six slots available, seven coin varieties in circulation:

Slot 1: P1 NGC Slot 2: P1 BSP old Slot 3: P5 NGC (nonagonal)
Slot 4: P5 BSP old Slot 5: P10 (both series share one slot) Slot 6: P20 NGC

P10 old and new share a slot — both bimetallic, only 0.5 apart. The two P5 varieties CANNOT share: 27.0 nickel-brass versus 25.0 nonagonal steel are nothing alike. Verify all figures with calipers and a scale on real coins before programming.

BOUGHT ITEMS (not fabricated)

Multi-coin acceptor, CH-926 class	1
Coin hopper, P1	1
Coin hopper, P5	1
SG90 micro servo	2
IR slot sensor (hopper outlet count)	2
M4 stainless fasteners	lot

REV	DATE	DESCRIPTION
A	_____	First issue — single swing diverter
B	_____	Swing diverter replaced by cascaded two-gate

EMATHRIX TECHNOLOGIES Katapatan Ave., Banay-Banay, Cabuyao City, Laguna DTI 7114652	
CUT LIST AND MATERIAL NOTES EMX-2026-WATERVENDO-01 COIN ROUTING REV B	
ALL DIMENSIONS IN MILLIMETRES	SHEET 4 OF 4 NOT TO SCALE